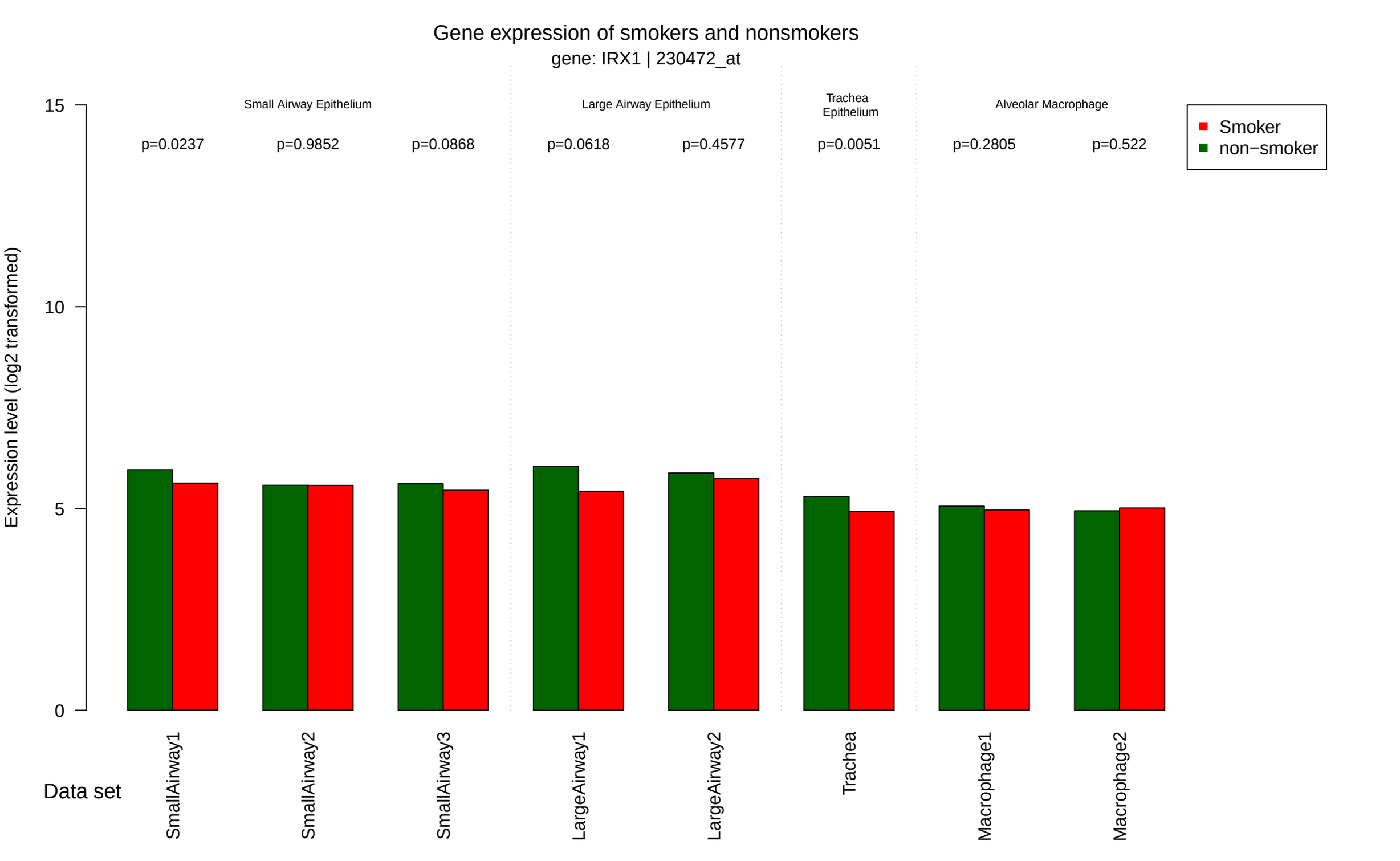
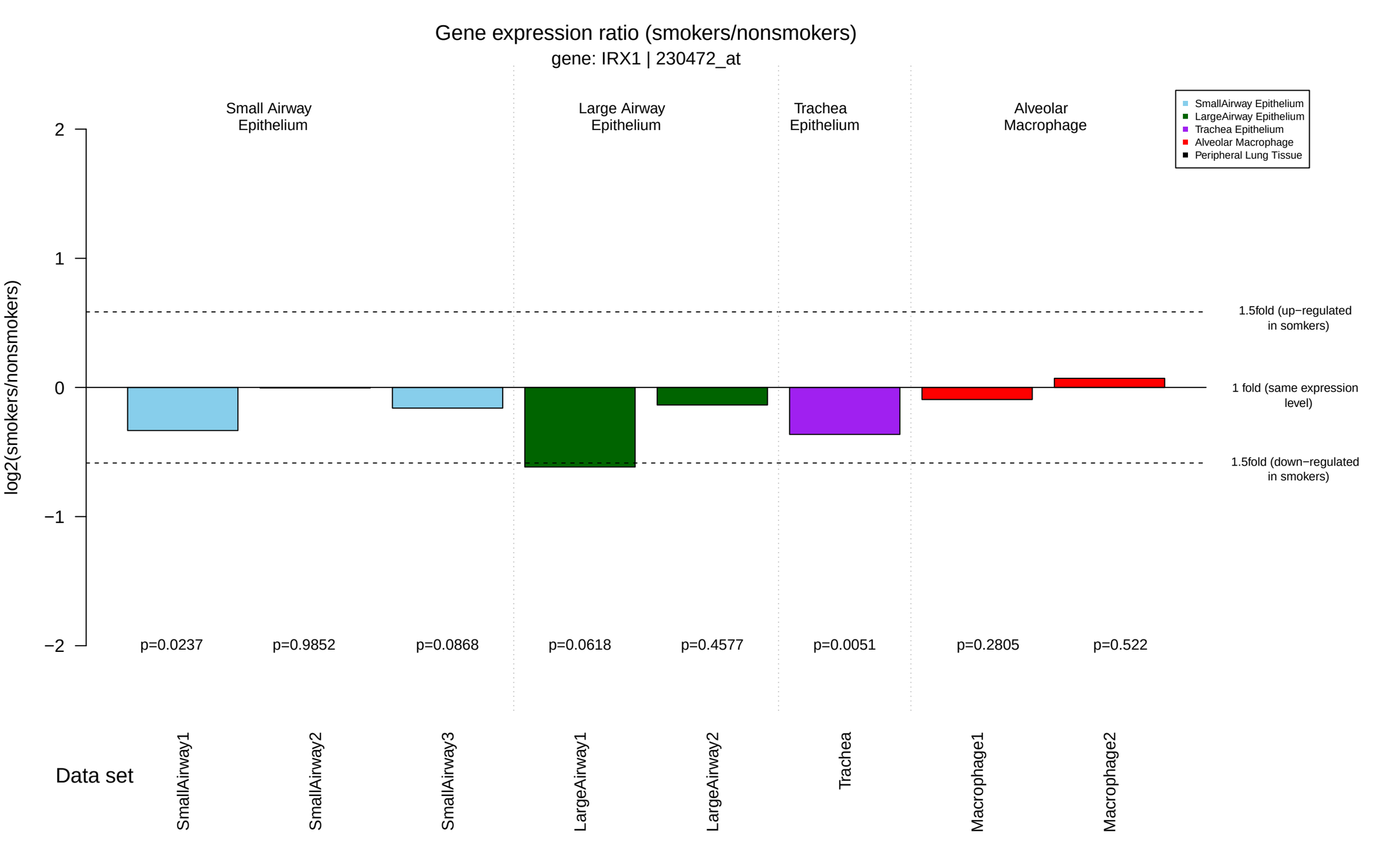
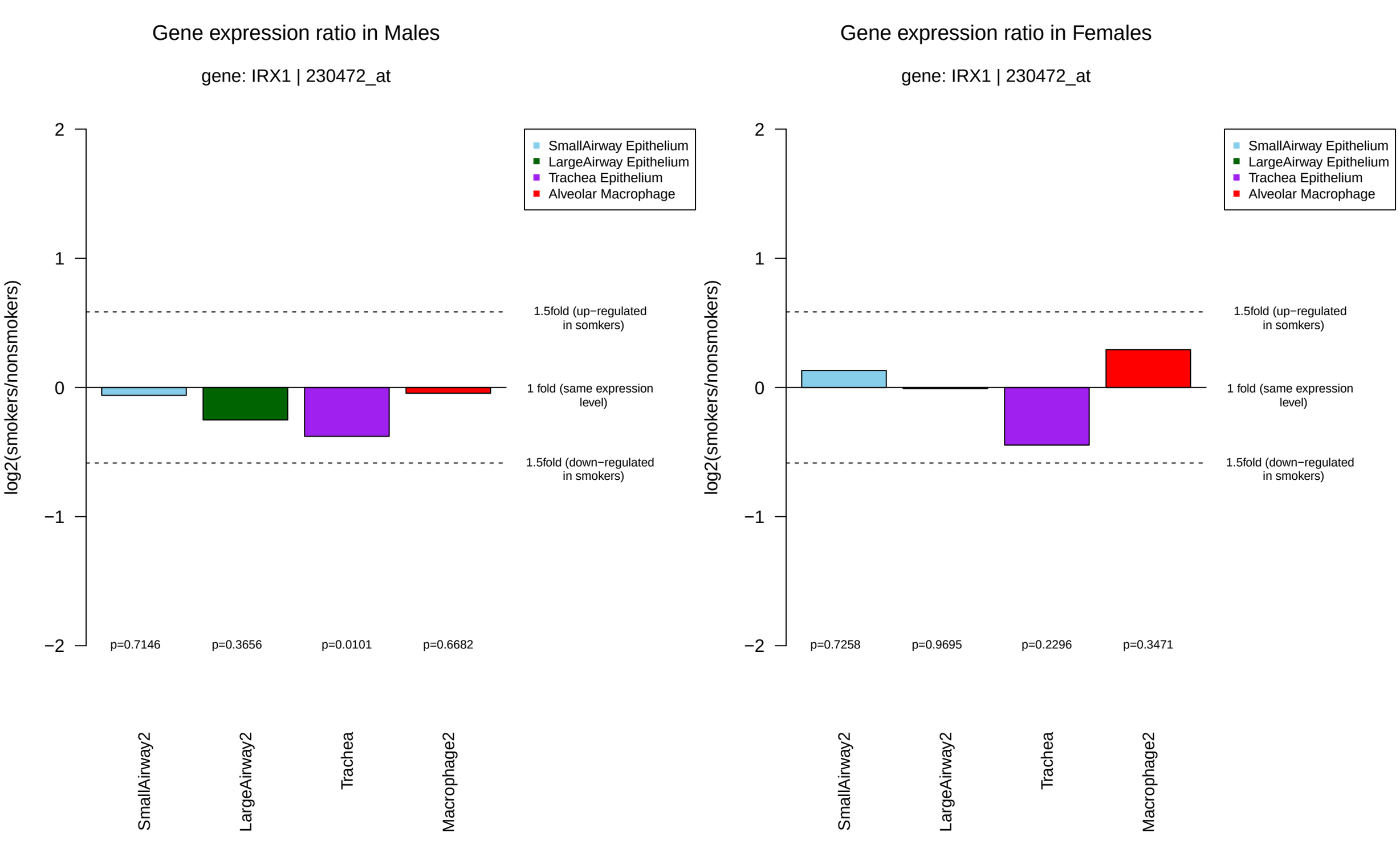
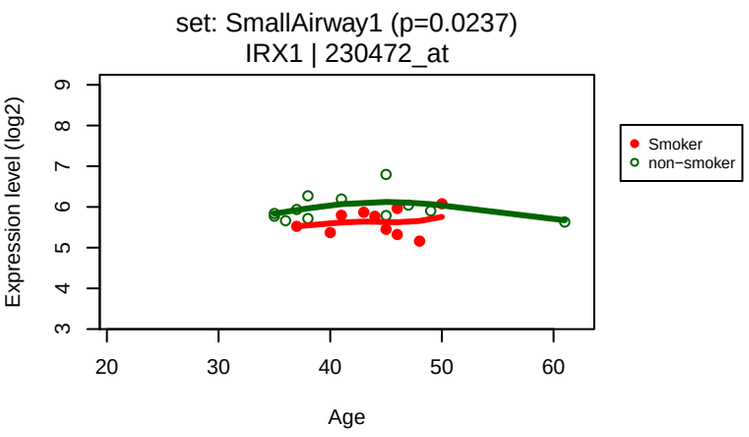
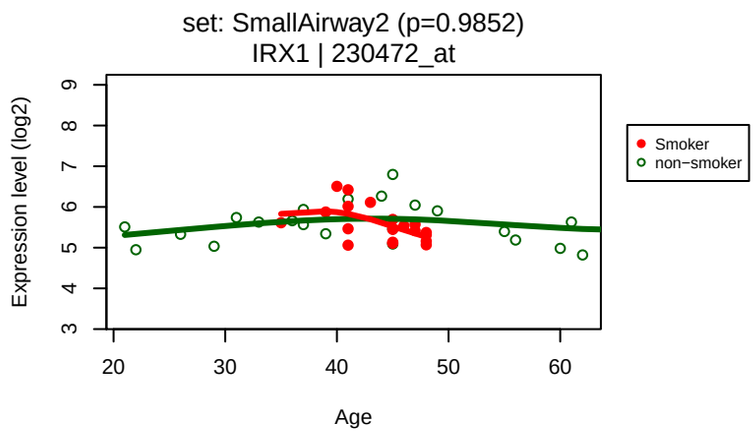
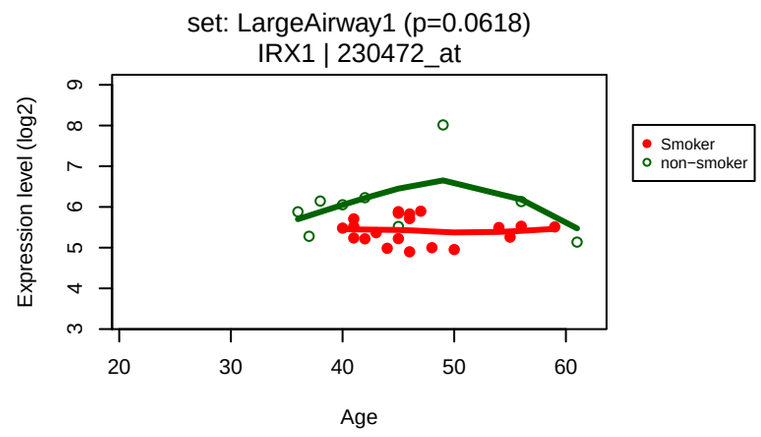

Gene expression of smokers and nonsmokers
gene: IRX1 | 230472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0237
p=0.9852
p=0.0868
p=0.0618
p=0.4577
p=0.0051
p=0.2805
p=0.522
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

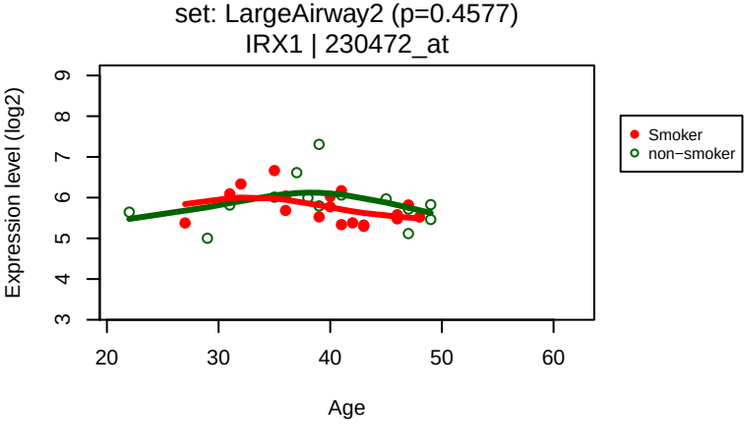

Gene expression ratio (smokers/nonsmokers)
gene: IRX1 | 230472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0237
p=0.9852
p=0.0868
p=0.0618
p=0.4577
p=0.0051
p=0.2805
p=0.522
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

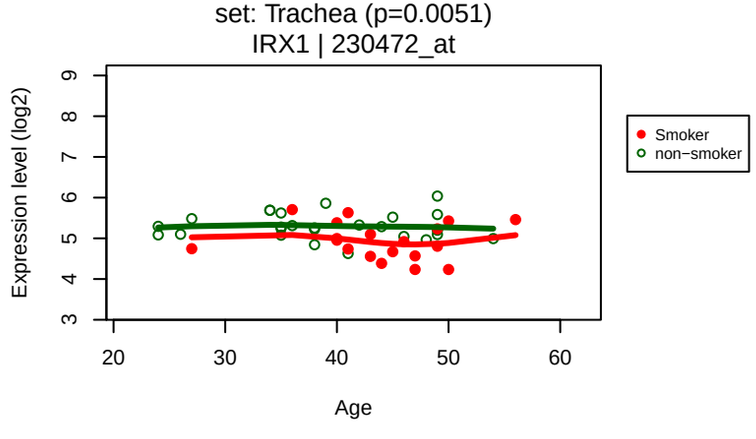

Gene expression ratio in Males
gene: IRX1 | 230472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7146
p=0.3656
p=0.0101
p=0.6682
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IRX1 | 230472_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7258
p=0.9695
p=0.2296
p=0.3471
SmallAirway2
LargeAirway2
Trachea
Macrophage2

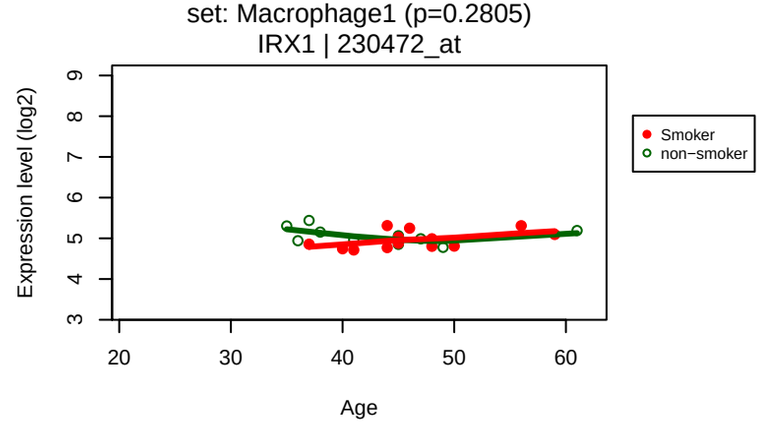

set: SmallAirway1 (p=0.0237)
IRX1 | 230472_at
Expression level (log2)
Age
Smoker
non−smoker

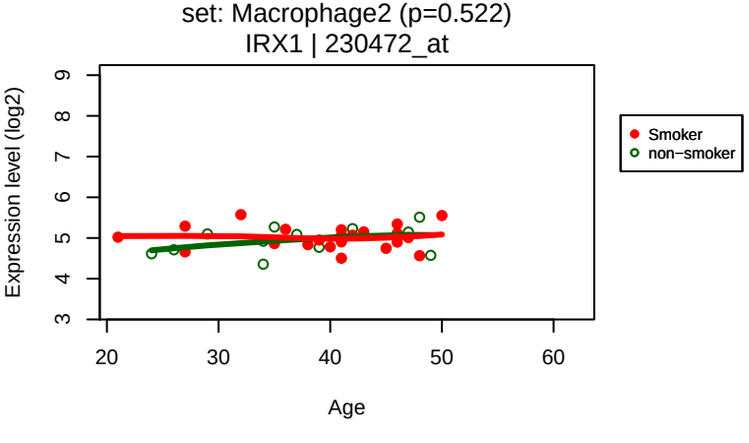

set: SmallAirway2 (p=0.9852)
IRX1 | 230472_at
Expression level (log2)
Age
Smoker
non−smoker

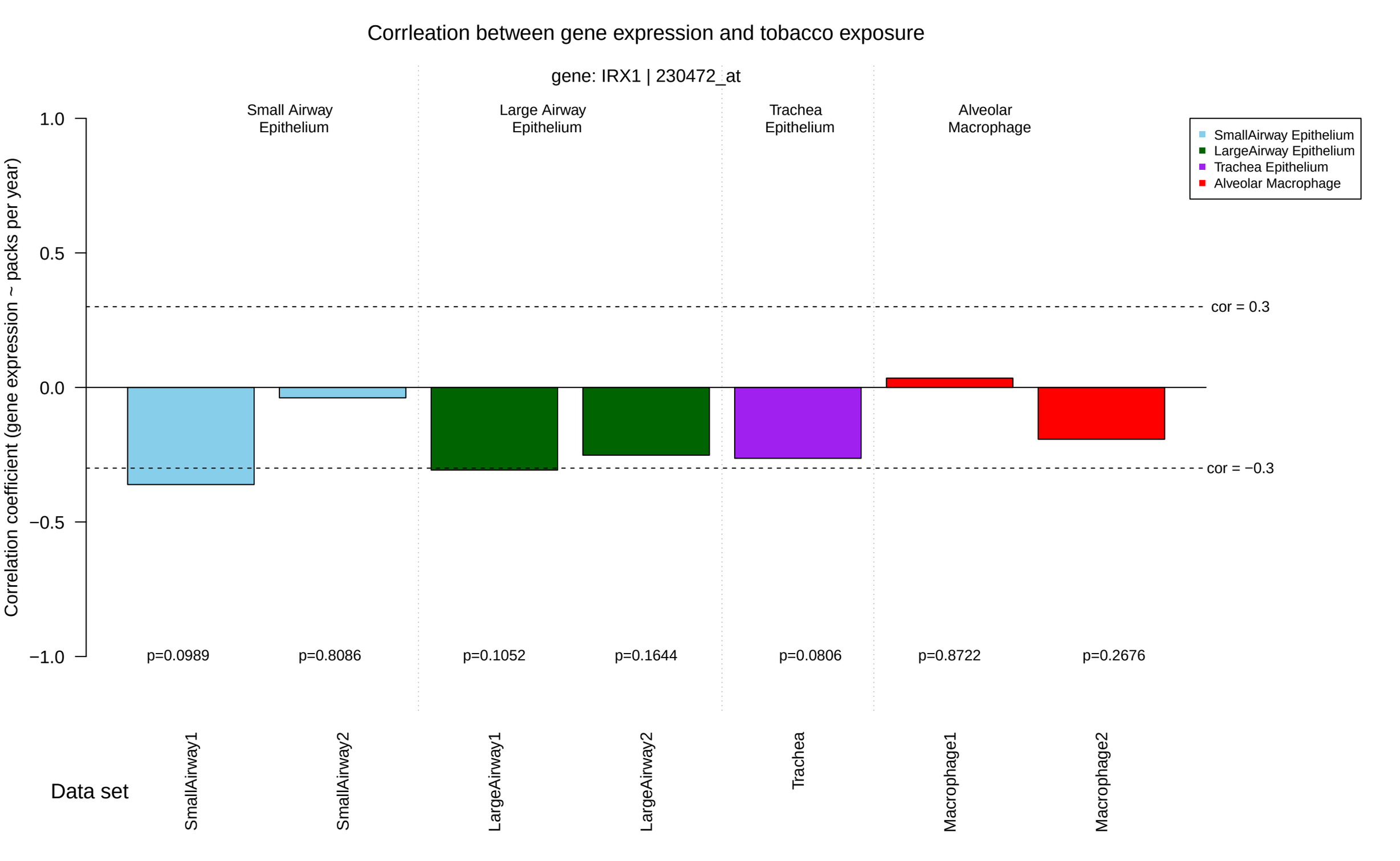

set: LargeAirway1 (p=0.0618)
IRX1 | 230472_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4577)
IRX1 | 230472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0051)
IRX1 | 230472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.2805)
IRX1 | 230472_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.522)
IRX1 | 230472_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IRX1 | 230472_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0989
p=0.8086
p=0.1052
p=0.1644
p=0.0806
p=0.8722
p=0.2676
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2